

```
1 grammar = {'S': [ (('NP', 'VP'), 1.0)], 'NP': [ (('Det', 'N'), 0.6), (('Name',), 0.4)], 'VP': [ (('V', 'NP'), 1.0)], 'Det': [ (('the',), 0.5), (('a',), 0.5)],
2      'N': [ (('cat',), 0.5), (('dog',), 0.5)], 'V': [ (('sees',), 0.5), (('likes',), 0.5)], 'Name': [ (('John',), 1.0)] }
3 sentence = input("Enter sentence: ").split()
4 def parse(symbol, pos):
5     if symbol not in grammar:
6         if pos < len(sentence) and symbol == sentence[pos]:
7             return 1, pos + 1
8         return 0, pos
9     best_prob = 0
10    best_pos = pos
11    for rule, probability in grammar[symbol]:
12        current = pos
13        prob = probability
14        success = True
15        for item in rule:
16            p, current = parse(item, current)
17            if p == 0:
18                success = False
19                break
20            prob *= p
21        if success and prob > best_prob:
22            best_prob = prob
23            best_pos = current
24    return best_prob, best_pos
25 probability, position = parse(symbol: 'S', pos: 0)
26 if position == len(sentence) and probability > 0:
27     print("Sentence accepted")
28     print("Probability:", probability)
29 else:
30     print("Sentence rejected")
```



```
4 def parse(symbol, pos): 2 usages
11     for rule, probability in grammar[symbol]:
12         current = pos
13         prob = probability
14         success = True
15         for item in rule:
16             p, current = parse(item, current)
17             if p == 0:
18                 success = False
19                 break
20             prob *= p
21         if success and prob > best_prob:
22             best_prob = prob
23             best_pos = current
24     return best_prob, best_pos
25 probability, position = parse(symbol: 'S', pos: 0)
26 if position == len(sentence) and probability > 0:
27     print("Sentence accepted")
28     print("Probability:", probability)
29 else:
30     print("Sentence rejected")
```

Run main

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↑ /Users/moshika/PyCharmMiscProject/.venv/bin/python /Users/moshika/PyCharmMiscProject/Exp 15.py
↓ Enter sentence: the cat sees a dog
⇌ Sentence accepted
⇌ Probability: 0.01125
⇌
⌵ Process finished with exit code 0
⌵
```

